

Cover letter – Human Genomics

Date: 10 October 2026

To: Professor Vasilis Vasiliou, Editor-in-Chief, Human Genomics

From: [CORRESPONDING AUTHOR NAME, DEGREE, AFFILIATION, POSTAL ADDRESS, EMAIL, TELEPHONE]

Re: Research manuscript, “Input-matched multi-signal colocalization clarifies immune-cell regulatory assignments at primary biliary cholangitis risk loci”

Dear Professor Vasiliou,

Please consider our Research manuscript, “Input-matched multi-signal colocalization clarifies immune-cell regulatory assignments at primary biliary cholangitis risk loci,” for publication in Human Genomics. The study addresses a recurring problem in post-GWAS interpretation: changes in colocalization labels can arise from variant-support and disease-statistic differences as well as from the move from single-causal to multi-signal inference.

We screened 6,923 PBC locus-gene-cell comparisons and applied source-matched multi-signal analysis to a frozen 642-comparison high-information subset. A four-arm A0/A1/A2/M design separated support restriction, disease-statistic matching and inferential procedure. Input matching explained part of the bidirectional reclassification, but stable H4-to-H3 and H3-to-H4 changes remained. Truth-known simulations exposed a trade-off between fewer false-H4 decisions and lower shared-signal recovery, with only one-third of iterations forming an evaluable signal pair. IL12RB2-NK and FCRL3 B-cell assignments received bounded cross-resource support, while the exact five-versus-five liver analysis supported lineage detectability without corrected PBC-specific enrichment.

The contribution is an evidence-governed analysis of regulatory-assignment reliability rather than a claim of new-gene discovery or universal method superiority. It fits Human Genomics through its integration of human genetic epidemiology, regulatory genomics, statistical genetics, single-cell molecular-QTL resources and transparent open-data reproducibility. The manuscript retains negative, model-sensitive and uninformative results and distinguishes statistical signal sharing from mechanism and mediation.

Code, frozen protocols, compact aggregate outputs, figure sources and machine-readable supplementary data are fixed in the public repository at <https://github.com/1209433622cz-maker/igan-open-regulatory-evidence/tree/r7b4a-human-genomics-interface-2026-10-10>. Third-party individual-level or licence-restricted files are represented by accession and provenance records rather than redistributed.

Before submission, the corresponding author must replace and confirm the following statements:

- [ALL AUTHORS APPROVE THIS VERSION AND ITS SUBMISSION.]
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- [FUNDING, AUTHOR CONTRIBUTIONS, ACKNOWLEDGEMENTS, APC ROUTE AND LICENCE ARE CONFIRMED.]

Thank you for considering this manuscript.

Sincerely,

[CORRESPONDING AUTHOR NAME AND SIGNATURE BLOCK]